

Return values and local state

Project: Python readiness portfolio

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Functions and arguments

Without opening the old lesson, explain one key idea from Functions and arguments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Loops and trace tables

Without opening the old lesson, explain one key idea from Loops and trace tables. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Expressions and operator order

Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace separate calls for Return values and local state. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Return values and local state. Proposed explanation: “Printing a value returns that value”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Use returns and keep local state local

Trace both calls and the final expression. If return total is removed from the function, does this final print complete? Would a separate print(total) outside the function have access to its local variable?

```
def twice(n):  
    total = 2 * n  
    return total  
  
first = twice(3)  
second = twice(5)  
  
print(first, second, first + second)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Separate returning from displaying

Using the correct worked definition, execute result = remaining(15, 6), then doubled = result * 2. State both stored values and what the script displays before a print statement is added.

My reasoning and evidence

6. W2 Trace separate calls

Using the correct definition, store `a = remaining(8, 3)`, `b = remaining(8, 8)`, `c = remaining(8, 3)`. State `a`, `b` and `c`. In a fresh script, explain why `print(spaces)` outside the function fails.

My reasoning and evidence

7. W3 Repair the returned value

Use the faulty printing version with `answer = remaining(11, 2)`. State the displayed number and the value stored in `answer`. Explain why `print(answer + 1)` fails, then repair the function and predict the repaired output.

My reasoning and evidence

8. W4 Build a reusable calculation independently

Each tray holds 5 counters. Define `stock(trays, loose)` to return the total using a local variable `total`. Assume nonnegative integer arguments. Store results for `(2, 3)`, `(0, 3)` and `(2, 0)`. Print the sum of the first two stored results. Predict that output and all three stored values. Call `(2, 3)` again to check fresh state.

My reasoning and evidence

9. W5 Locate a statement that never runs

A function body has spaces = capacity - used, then return spaces, then print("done"), all at the same indentation. What does answer = remaining(4, 4) store and display? Explain whether the final print can run.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled return values and local state evidence sheet to Python readiness portfolio. Show how you can use a returned value in a later calculation. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Return values and local state

Return values survive their calls. A function returns a local calculation that the caller stores.

Python 3: `def extra(a,b): result=a+b; return result. Then first=extra(2,5); second=extra(4,1); print(first,second).`

Return values and local state

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Trace first call

Its local result is 7; return assigns 7 to first.



2. Trace second call

Its new local result is 5; return assigns 5 to second.



3. Read caller values

The final print displays 7 5; the second call does not overwrite first.

CHANGE ONE THING

Add `print(result)` after the calls in a fresh script with no global result.

Explain the changed result using the matching lesson.

Trace the two returned values and explain why first stays seven. What happens if the caller tries to print result directly?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.